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Torricelli's Infinitely Long Solid and Its Philosophical Reception in the Seventeenth Century

By Paolo Mancosu and Ezio Vailati***

IN 1641 EVANGELISTA TORRICELLI discovered that a certain solid of infinite length, which he called the "acute hyperbolic solid," has a finite volume. This paradoxical result created considerable interest in mathematical and philosophical circles around the third quarter of the seventeenth century. The result seemed so counterintuitive and astonishing that at first some of the leading mathematicians thought it impossible; even eighty years later Bernard de Fontenelle commented, "One apparently expected, and should have expected, to find [Torricelli's solid] infinite" in volume.[1] Some important philosophers, too, were sufficiently struck by the oddity and the implications of Torricelli's result to comment on it. Of course not all reacted in the same way, and some were more forthcoming than others. But by and large philosophical responses tended to cluster around three main topics, all of considerable importance in the seventeenth century. First, because of its infinite length Torricelli's hyperbolic solid stretched the intuitive universe of geometrical figures constituting the intellectual background of the early modern geometer and, consequently, raised the issue of the nature and ontological status of mathematical objects. Second, Torricelli's result called into question traditional philosophical assumptions about infinity and, in so doing, brought to the foreground the role of "natural light" in geometry and, by implication, the position of mathematics in the web of knowledge. Finally, by providing a nontrivial piece of knowledge about infinity, his theorem highlighted the issues of our knowledge and our idea of infinity. However, the role of Torricelli's result in occasioning philosophical discussions about these three issues has gone virtually unnoticed. This article studies Torricelli's theorem and the philosophical problems and discussions it generated.

TORRICELLI'S RESULT

When Torricelli was given the chair of mathematics in Florence in 1642 as the successor of Galileo he was virtually unknown to the geometrical community. By

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[1] Bernard de Fontenelle, *Élémens de la géométrie de l'infini,* in *Suite des mémoires de l'Académie Royale des Sciences,* 1725 (Paris, 1727), p. 457. Fontenelle does not mention Torricelli by name but only describes the acute hyperbolic solid. All translations are ours unless otherwise stated.

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the time his *Opera geometrica,* the only work he published in his lifetime, appeared in 1644, Torricelli had become one of the most acclaimed geometers in Europe.[2] This change of status occurred in 1643, when Torricelli communicated several discoveries he had made to the French geometers. The most important discovery had to do with the study of a remarkable geometric solid. If one revolves a branch of the Apollonian hyperbola ($xy = a^2$) around one of the asymptotes (see Fig. 2 below), one obtains a solid that is infinitely long in the direction of the axis of revolution. By cutting the solid so obtained by a plane perpendicular to the axis of revolution, one obtains a solid of infinite length whose volume is finite. This result was to constitute the core of Torricelli's work *De solido hyperbolico acuto,* published in *Opera geometrica.*

Torricelli's discovery dated back to 1641, when he communicated the result to his friend and patron Bonaventura Cavalieri. Although Torricelli's letter is lost, we do have Cavalieri's reply.

> I received your letter while in bed with fever and gout . . . but in spite of my illness, I enjoyed the savory fruits of your mind, since I found infinitely admirable that infinitely long hyperbolic solid which is equal to a body finite in all the three dimensions. And having spoken about it to some of my philosophy students, they agreed that it seemed truly marvelous and extraordinary that that could be.[3]

In 1643 Torricelli sent a letter to Jean-François Niceron in Paris, attaching a single sheet that communicated his geometrical discoveries (but not the proofs) to the French mathematicians. Marin Mersenne was immediately informed, and he in turn told Gilles Personne de Roberval, Pierre de Fermat, and others. Torricelli's celebrity was immediate. Michelangelo Ricci wrote to him: "Father Niceron writes me that all the excellent men of that kingdom desire to see your works and that the sheet of propositions I sent them is passing through everyone's hands with great praise of your beautiful discoveries [*invenzioni*]."[4]

The fourteenth proposition of Torricelli's letter (on the infinitely long solid) was the most astonishing one. Even if Torricelli could not claim to have broken new ground in his other results, the cubature of the infinitely long solid brought him to the forefront of European research in geometry. Roberval wrote to

[2] Evangelista Torricelli, *Opera geometrica* (Florence, 1644), in *Opere di Evangelista Torricelli,* ed. Gino Loria and Giuseppe Vassura, 4 vols. in 5 parts (Faenza: Stabilimento Tipo-Litografico Montanari, 1919–1944) (hereafter **Torricelli, *Opere***). The *Opera geometrica* was completely translated into Italian by Luigi Belloni in *Opere scelte di Evangelista Torricelli* (Turin: UTET, 1975). Parts of the work *De solido hyperbolico acuto,* to which we will often refer in this paper, were translated into English in D. J. Struik, *A Source Book in Mathematics, 1200–1800* (Princeton, N.J.: Princeton Univ. Press, 1986), pp. 227–231.

[3] Bonaventura Cavalieri to Evangelista Torricelli, 17 Dec. 1641, *Opere,* Vol. III, p. 65. Cavalieri's admiration for the infinitely long solid was not a passing one, for he mentions the result in his last work, *Exercitationes geometricae sex* (Bologna, 1647), p. 2: "Corpus (quod ipse [Torricellius] vocat, Solidum acutum hyperbolicum) infinitae longitudinis, geometricae insolens, sed tamen admirabile . . . per indivisibilia tamen curva . . . mensura subiecit." Cavalieri himself gave a construction of an infinitely long solid of finite volume in *Exercitationes,* prop. 38, p. 536. See the free translation by Evelyn Walker, *A Study of the* Traité des indivisibles *of Gilles Personne de Roberval* (New York: Teachers College, 1932), pp. 68–69.

[4] Torricelli to J.-F. Niceron, [May or June 1643], in *Correspondance du P. Marin Mersenne,* Vol. XII, ed. Cornélis de Waard and Bernard Rochot (Paris: Éditions du Centre National de la Recherche Scientifique, 1972), p. 206; and Michelangelo Ricci to Torricelli, [May or June/1643], in *Le opere dei discepoli di Galileo Galilei,* Vol. I: *Carteggio, 1642–1648,* eds. P. Galluzzi and M. Torrini (Florence: Giunti Barbera, 1975), p. 63.

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Mersenne that he considered Torricelli's result very elegant and then added his own proof of the statement.[5]

A complete account of Torricelli's work on the infinitely long solid (which contained much more than the cubature of the solid) was presented in *De solido hyperbolico acuto*.[6] The techniques that led to the determination of the volume were provided by Cavalieri's theory of indivisibles, which Torricelli had recently embraced with the enthusiasm of the neophyte. Although there were several different versions of the theory of indivisibles (Cavalieri distinguished between two approaches called, respectively, collective and distributive, and Torricelli professed his own brand of indivisibilism), the main intuition behind it was to consider a plane figure as characterized by its plane indivisibles (i.e., its segments) and a solid as characterized by its plane sections. Then two plane figures (solids) of the same height have the same areas (volumes) if and only if the class of indivisibles characterizing the first figure is equal to the class of indivisibles characterizing the second figure.[7]

Torricelli's commitment to the indivisibilist method is evidenced by the favorable comparison he drew between it and the traditional indirect method of proof:

> As for the method of demonstration, we shall prove a simple notable theorem in two ways, namely, with indivisibles and in the manner of the ancients. And this although, to tell the truth, it has been discovered with the geometry of indivisibles, which is a truly scientific method of demonstration which is direct, and so to say, natural. I feel pity for the ancient Geometry which not knowing or not allowing Indivisibles discovered so few truths in the study of the measure of solids, that a frightening paucity of ideas has continued until our times.[8]

It was important to defend the method of indivisibles because Torricelli's *Opera geometrica* appeared at a critical moment. The method, used and explained by Cavalieri in his *Geometria continuorum nova ratione promota* in 1635, had been attacked in 1641 by Paul Guldin in the second volume of his *Centrobaryca* as being nondemonstrative.[9] Torricelli, therefore, emphasized not only the rigor, generality, and directness of the method but also its productivity—which, he claimed, made the old geometry look like a poor relation of the new one.

Torricelli did more than cleverly apply Cavalieri's techniques. For one thing, he vastly increased the range of applications of the indivisibilist method by extensively using curved indivisibles:

> Our method, which we are about to use in the aforementioned theorem, will proceed with curved indivisibles without following the example of any predecessor but not

[5] Roberval to Mersenne, [July 1643], in Mersenne, *Correspondance,* Vol. XII, p. 257. The statement of the theorem Torricelli sent to the French mathematicians is equivalent to the one he proved in "De solido." For the significance of the discovery see Jean Itard, "La lettre de Torricelli à Roberval d'octobre 1643," *Revue d'Histoire des Science,* 1975, *28*:113–124, on p. 122.

[6] We cannot study the technical developments initiated by Torricelli's theorem in this article. For Torricelli's further work on infinite hyperbolas see Ettore Bortolotti, "La memoria *De infinitis hyperbolis* di Torricelli," *Archeion,* 1925, *6*:139–152; and Bortolotti, "La scoperta e le successive generalizzazioni di un teorema fondamentale di calcolo integrale," *Archeion,* 1924, *5*:205–227, esp. pp. 212–213, on the acute hyperbolic solid.

[7] Our characterization is intentionally incomplete. For details see Enrico Giusti, *Bonaventura Cavalieri and the Theory of Indivisibles* (Rome: Cremonese, 1980); and Kirsti Andersen, "Cavalieri's Method of Indivisibles," *Archive for History of Exact Sciences,* 1985, *31*:291–367.

[8] Evangelista Torricelli, *De solido hyperbolico acuto,* in *Opere,* Vol. I, p. 173.

[9] See Giusti, *Cavalieri and the Theory of Indivisibles* (cit. n. 7), esp. pp. 55–65.

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> without the previous approval of Geometricians. . . . Curved indivisibles adequate for these demonstrations are peripheries of circles in plane figures, and spherical, cylindrical, and conic surfaces in solid figures. They have the advantage of fitting perfectly and having, so to say, a thickness which is always equal and uniform.[10]

Furthermore, as the quotation indicates, he conceived of curved indivisibles as having a thickness; Cavalieri's indivisibles have a lower dimension than the figure they characterize, and therefore one cannot speak of a thickness of indivisibles in his account. Finally, Torricelli substituted for the complexity of Cavalieri's approach (in which the geometric figure and the class of its indivisibles were considered as two separate magnitudes) a simple identification of the figure with its characterizing indivisibles.[11] This can be illustrated by a simple example that Torricelli presented as a "warm-up" exercise on curved indivisibles before he proved the main theorems of *De solido hyperbolico acuto*. It is a proof of the Archimedean proposition on the measurement of the circle: the area of a circle is equal to the area of a right triangle whose legs are equal to the radius and the circumference of the circle (see Fig. 1).

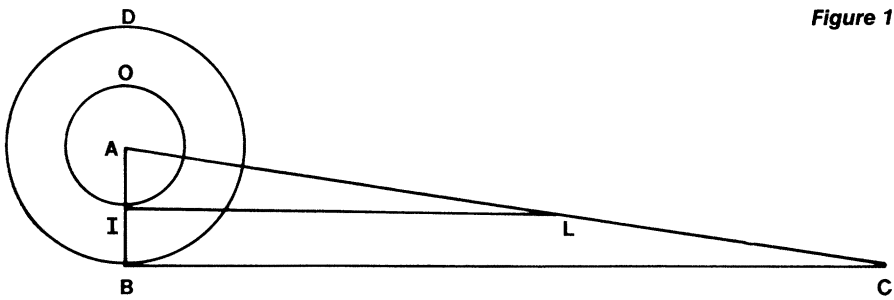

Figure 1

> Draw the circle *BDB* with radius *AB*. Consider an arbitrarily chosen point *I* on *AB*. Let *BC* equal the circumference *BDB*. If one produces a second circle with center *A* and radius *AI* one obtains the following proportions:

$$\text{circumference } BDB : \text{circumference } IOI = AB:AI$$
$$= BC:IL$$

Hence

$$\text{circumference } BDB:BC = \text{circumference } IOI:IL.$$

Consequently

$$\text{circumference } IOI = IL.$$

Since this is true for any arbitrarily chosen point *I* on *AB*, Torricelli concluded: "Therefore, all the peripheries taken together are equal to all the straight lines

[10] Torricelli, *De solido,* in *Opere,* Vol. I, p. 174. Torricelli seems concerned to defend his originality even though Cavalieri had mentioned to him in 1641 that he had already made use of curved indivisibles in order to compare spirals with parabolas. In fact there are a few theorems in the *Geometria* that make use of curved indivisibles.

[11] Kirsti Andersen has shown that many mathematicians learned the techniques of indivisibles through Torricelli's version rather than through the original texts of Cavalieri. This accounts, in her opinion, for the common misunderstandings about the nature of Cavalieri's method. See Kirsti Andersen, "The Method of Indivisibles: Changing Understandings," in *300 Jahre "Nova Methodus" von G. W. Leibniz (1684–1984),* ed. Albert Heinekamp (*Studia Leibnitiana,* Sonderheft 14) (Stuttgart: Steiner Verlag, 1986), pp. 14–25.

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taken together. That is, the circle *BD* will be equal to the triangle *ABC*."[12] In other words, Torricelli identified the circle with the class of its curved indivisibles. In measuring the circle we dealt with plane curved indivisibles; in the proof of the finite volume of the infinitely long solid one needs to use solid curved indivisibles.

Before considering Torricelli's proof it is important to look at his definition of the hyperbolic solid and his statement of the theorem, because, as we shall see, they bring out the infinitistic nature of the result. For this reason both deserve to be quoted in full.

> DEFINITION. *If one rotates a hyperbola around an asymptote, as around an axis, one generates a solid infinite in length* [longitudine infinitum] *in the direction of the axis which we call an acute hyperbolic solid.*

> THEOREM. *An acute hyperbolic solid, infinitely long* [infinite longum], *cut by a plane [perpendicular] to the axis, together with a cylinder of the same base, is equal to that right cylinder of which the base is the latus transversum of the hyperbola (that is, the diameter of the hyperbola), and of which the altitude is equal to the radius of the base of this acute body.*[13]

Thus we want to show that the solid described in the statement of the theorem composed by the cylinder *FEDC* (see Fig. 2) and the solid generated by the rotation of the segment of the hyperbola with lowest point *D* around the axis *AB* are together equal to the cylinder whose height is *CA* and whose base is the circle of diameter $HA = 2AS$.

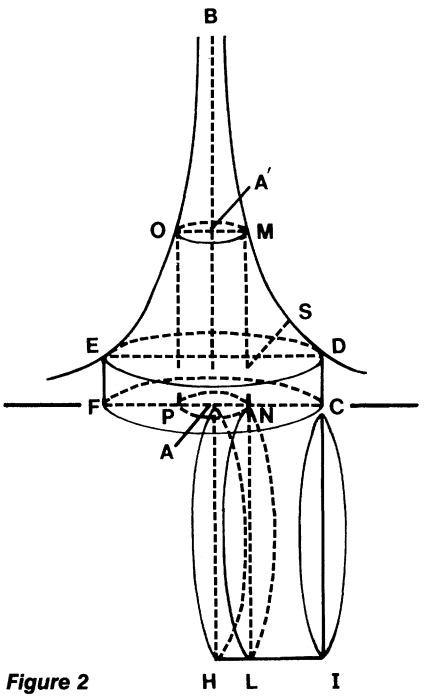

Figure 2

[12] Evangelista Torricelli, *De infinitis parabolis,* in *Opere,* Vol. I, p. 321.
[13] Torricelli, *De solido,* in *Opere,* Vol. I, pp. 190 (definition), 193 (theorem).

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The proof by indivisibles

The proof is preceded by five preliminary lemmas whose demonstrations do not make use of indivisibles. The first four build up to lemma 5, which is explicitly used in the proof of the theorem. Lemma 5 asserts that the lateral surface of the inscribed cylinder *POMN* (see Fig. 2), whose axis lies on *AB* and whose bases are the circles with diameters *OM* and *PN*, is equal to the area of a circle whose radius is *AS*, the semidiameter of the hyperbola, and that this holds for any inscribed cylinder of that form.

The idea of the proof is very simple. We have to show that the curved indivisibles of the infinitely long solid described in the statement of the theorem are equal to the curved indivisibles of the cylinder *ACIH*. The infinitely long solid is thought to be made up of all its cylindrical indivisibles, that is, all lateral surfaces of the type *POMN*. The cylinder *ACIH* is thought of as being made up of all its circular indivisibles, that is, the cross-sections whose diameter is *AH*. Some arbitrarily chosen point *N* determines an indivisible in the infinitely long solid and an associated indivisible in the cylinder *ACIH*. But each indivisible of the infinitely long solid (i.e., a lateral surface of the type *POMN*) is equal in area (by lemma 5) to the circle whose radius is *AS*, that is (since by construction $AH = 2AS$), to an indivisible of *ACIH*. Therefore, since the indivisibles of the two figures are equal, by the fundamental principle of the theory of indivisibles, the volumes of the two figures will be equal too. Hence the infinitely long solid is equal to the finite cylinder whose base is the circle with diameter *AH* and whose height is *AC*. This concludes the proof.

In the scholium following the proof Torricelli made two interesting remarks:

> It may seem incredible that although this solid has an infinite length [*infinitam longitudinem*], nevertheless none of the cylindrical surfaces we considered has an infinite length, but all of them are finite, as will be clear to everyone who is even modestly familiar with the doctrine of Conics.
>
> I consider the previous theorem sufficiently clear in itself, and more than adequately confirmed by the examples proposed at the beginning of the book. However, in order to satisfy also the reader who is scarcely a friend of indivisibles, I shall repeat its demonstration at the end of the work with the usual demonstrative method of ancient Geometers which, although longer, in my opinion is not for that more certain.[14]

The first comment initially seems confusing. In order to fill the solid with the lateral surfaces of the inscribed cylinders, one has to consider the limit case, in which the lateral surface of the innermost cylinder degenerates into an infinitely long line, the axis of the solid. Thus when Torricelli said that none of the lateral surfaces has an infinite length, he meant that none of them has an infinite area. This is confirmed in a note published posthumously: "What Galileo says of a point equal to a line is true, and in our hyperbolic solid it is true that the infinitely long axis is equal to a circle [*circolo quanto*]."[15] We should note that in the proof Torricelli took for granted that the equality of the lateral surfaces with a specified

[14] *Ibid.*, p. 194.

[15] Torricelli, "De infinitis parabolis," in *Opere*, Vol. I, p. 321. The reference is to the paradox of the "soup dish" presented by Galileo in the first day of the *Two New Sciences*, trans. Stillman Drake (Madison: Univ. Wisconsin Press, 1974), pp. 37–38.

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circle holds also when $N = A$, that is, when the lateral surface degenerates into a straight line. This is not, as he implied, a consequence of the theorem but an assumption that is necessary for the completeness of the proof. To our knowledge, none of his contemporaries remarked on this aspect of the proof.

The second comment concerns the necessity of proving the result by the method of exhaustion. Torricelli declared himself satisfied by the proof just given, but he added that he would offer a proof by the "usual demonstrative method of ancient Geometers" for those who did not like indivisibles. Behind this conciliatory and perhaps condescending attitude we can perceive his cautiousness. Because the method of indivisibles was under attack, it was safer to publish both proofs. Torricelli's strategy is reminiscent of the caution shown by Archimedes who, in his *Quadrature of the Parabola,* approaching Dositheus for the first time, had sent two proofs of his results, the first mechanical and the second purely geometrical. Although Archimedes expressed no doubts as to the validity of the first proof, he apparently preferred to provide a purely geometrical proof in order to avoid criticism on formal grounds. In introducing the proof by exhaustion, Torricelli remarked that it would seem an impossible task to inscribe another figure in a figure of infinite length, or to circumscribe another figure around such a figure. However, after praising Roberval's proof by exhaustion, he proceeded to give his own.

The proof by exhaustion

The result to be proved is the same as before. We want to show that the solid described in the statement of the theorem, call it H, composed by the cylinder $FEDC$ (see Fig. 2) and the solid generated by the rotation of the segment of the hyperbola with lowest point D around the axis AB together are equal to the cylinder whose height is CA and whose base is the circle of diameter $HA = 2AS$. Since the proof is by exhaustion, we will approximate the solids with figures of the same dimension, that is, with finite solids. The two main lemmas needed for Torricelli's theorem are lemmas 6 and 7.

> LEMMA 6: *Consider the solid described by the rotation of the mixtilinear figure $MNCD$ around AB. The hollow solid described is equal in volume to the cylinder whose height is NC and whose base is the circle with diameter NL. This is true for any arbitrarily chosen point N on AC different from A.*
>
> LEMMA 7: *Consider the cylinder generated by the rotation of the rectangle $ANMA'$ around AB. The volume of this cylinder is half the volume of the cylinder whose height is AN and whose base is the circle with diameter AH. This also holds true for any arbitrarily chosen point N on AC different from A.*

We are now ready for the main proof. Let us denote by C the cylinder $HACI$. To prove $V(H) = V(C)$, where V stands for volume, one proceeds, as usual, by contradiction.

> Assume $V(H) < V(C)$. Then H is equal to a piece of C, let us say to the cylinder $LNCI$. Extend NL to meet the hyperbola in M. Then the solid generated by the rotation of the mixtilinear figure $MNCD$ around the axis AB

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is, by lemma 6, equal to the cylinder *LNCI;* and this is true for any point N on AC different from A. Thus, since the hollow solid generated is part of H, we have $V(C) < V(H)$, a contradiction.

Assume then $V(C) < V(H)$. Since the value of C is finite, it will be equal to some finite segment of H, say *FEOMDC*. Call this segment W. Thus $V(W) = V(C)$. W is the sum of the hollow solid T, generated by the rotation of the mixtilinear figure *MNCD*, and the cylinder Z, obtained by the rotation of the rectangle AM. But $2V(Z) = V(\text{cylinder } ANLH)$ (by lemma 7). Hence

$$\begin{aligned} V(W) &= V(Z) + V(T) \\ &= \tfrac{1}{2}V(\text{cyl. } ANLH) + V(T) \text{ (by lemma 7)} \\ &= \tfrac{1}{2}V(\text{cyl. } ANLH) + V(\text{cyl. } LNCI) \text{ (by lemma 6)} \\ &< V(\text{cyl. } ANLH) + V(\text{cyl. } LNCI) = V(C). \end{aligned}$$

Consequently, $V(W) \neq V(C)$, contrary to the assumption. Hence, $V(C)$ cannot be less than $V(H)$. Since both $V(H) > V(C)$ and $V(H) < V(C)$ are false, $V(H) = V(C)$. This concludes the proof.

THE PHILOSOPHICAL RELEVANCE OF TORRICELLI'S RESULT

To understand the philosophical reactions that Torricelli's result generated, it is important to note its infinitist nature. The theorem, as it stands, is about the proportion between two figures: an infinitely long solid and a cylinder. Thus the notion of actually infinite length is present in the very statement of the theorem. Moreover, the proofs themselves, as they stand, make sense only if the hyperbolic solid is given as infinitely long *in actu*. For we are comparing the volume of a fixed determinate solid with that of a cylinder, and there is no indication that the volume of the hyperbolic solid is thought of as the limit of a succession of volumes converging to that of the cylinder. Furthermore, Torricelli always referred to the hyperbolic solid as being infinitely long; and even Cavalieri, who was otherwise very careful in using the term *indefinite* rather than the term *infinite*, referred to it in the same way.[16] The most striking feature of Torricelli's result, then, is that the acute hyperbolic solid, although finite in volume, is not merely potentially but *actually* infinite in length. This gives rise to three philosophical issues, two of which can be characterized as epistemological and one as ontological.

The first epistemological issue is that of our knowledge of infinity, or, in seventeenth-century jargon, that of the nature of our idea of infinity. Broadly speaking, two positions were prominent around the middle of the century: that we have merely a negative idea of infinity, and that we have a positive one. The former position is traditionally associated with empiricist epistemology (e.g., Pierre Gassendi and Thomas Hobbes) and essentially reduces to the view that when we say of something that it is infinite, all we mean, or ought to mean, is that we are unable to conceive its limits. In other words, we call something "infinite" just when it is epistemically possible that it is boundless. Properly speaking, then, our talk about infinity is nothing but talk about our own inability and does not involve any sort of knowledge of infinity per se. The latter position,

[16] For Cavalieri's remarks see notes 3 and 17.

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in contrast, holds that we do have some knowledge, however minimal, of infinity per se. As Hobbes's reaction will show, Torricelli's cubature makes problematic the view that we have merely a negative idea of infinity, for it certainly seems to provide us with some positive and nontrivial knowledge about infinity.

The second epistemological issue arises from the combination of the paradoxical nature of the theorem and certain assumptions quite widespread around the third quarter of the seventeenth century. Torricelli himself remarked that his result was both "new and, as it were, paradoxical." The reason is that "if one proposes to consider a solid, or a plane figure, infinitely extended, everybody immediately thinks that such a figure must be of infinite size." But Torricelli's theorem shows that this intuitive belief is wrong. If we are to judge from the reactions of Torricelli's contemporaries, he was not alone in finding the result paradoxical. Cavalieri was shocked; he wrote to Torricelli, "I thank you for the truly divine demonstration of the acute hyperbolic solid; and I do not know how you fished out its measure so easily in the infinite depth of that solid. For, it seems infinitely long [*lungo*] to me because the plane figure [*spazio*] which generates it appears infinitely long to me, and each of its parts generates part of the solid." Nor was he the only one to be surprised. In his *Racconto* Torricelli accused Roberval of not having believed the result at first and even of having doubted its feasibility: "Mr. Roberval considered this proposition false and impossible; actually, when Father Mersenne came by here he told me that Roberval, having thought about it for some time, wrote I don't know what demonstrations or speeches [*discorsi*] in order to prove that my proposition was absurd and impossible."[17]

Many a philosopher held that geometry provides us with indisputable knowledge and that all knowledge involves a set of self-evident truths known by natural light. But there are good intuitive grounds for holding that an infinitely long solid cannot have a finite volume. Furthermore, such a conclusion seems clearly entailed by the age-old and, according to Isaac Barrow, widespread Aristotelian dictum that there is no proportion between the finite and the infinite.[18] We have, then, all the ingredients for a contradiction, which can be dispelled by abandoning some of the assumptions, by reinterpreting them, or by reinterpreting Torricelli's result. We shall see that Barrow combined the first and second alternatives and Hobbes took the third.

In addition to the two epistemological issues just mentioned, because of its infinite length the acute hyperbolic solid also raises the issue of the ontological status of geometrical entities. Torricelli himself pointed out that "in school tracts of geometry one finds figures limited in every side, and among all the solids of which ancient and modern authors have determined the measures with much effort, none, as far as I know, has an infinite extension."[19] He seemed aware that

[17] Evangelista Torricelli, *De dimensione parabolae,* in *Opere,* Vol. I, p. 94; Torricelli, *De solido, ibid.,* p. 173; Cavalieri to Torricelli, 7 Jan. 1642, in *Opere dei discepoli,* ed. Galluzzi and Torrini (cit. n. 4), pp. 1–2 (although its general tone is clear, we have been unable to make complete sense of Cavalieri's statement); and Torricelli, *Racconto di alcuni problemi proposti e passati scambievolmente tra gli matematici di Francia et il Torricelli ne i quattro anni prossimamente passati,* in *Opere,* Vol. III, pp. 7–32, on p. 13.

[18] Aristotle's dictum occurs in *De coelo* 1.6, 274a.

[19] Torricelli, *De solido,* in *Opere,* Vol. I, p. 173. Torricelli was not quite right. In criticizing Aristotle's point that an infinite body must be infinite in all three dimensions, Nicole Oresme had not only

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he was opening new vistas for geometry by challenging a number of informal assumptions that ruled and structured geometrical discourse. The notion of an infinitely long solid is perfectly consistent with the definition of a solid given in Euclid ("a solid is that which has length, breadth, and depth"), but it was apparently at odds with the intuitive universe of geometrical entities constituting the background of the early seventeenth-century geometer. Evidence that this was so is provided by Mersenne's remark in the introduction to his *Universae geometriae mixtaeque mathematicae synopsis* (1644). Commenting on Torricelli's result, he expressed serious concern regarding the propriety of using the word *space* to denote that "which is not closed [*clauditur*] on every side if not, perhaps, at infinity."[20] This quotation reveals that we are witnessing the stretching of some basic intuitive geometrical notions: Mersenne is confronting the idea that figures and solids need not be bounded on every side.

But Torricelli's solid, besides generating the conceptual problem of admitting a solid not bounded on every side (a problem that, we shall see, Hobbes addressed as well), dramatizes the further issue of the nature of geometrical entities. This topic was touched upon by Torricelli in the introduction to *De dimensione parabolae,* where he addressed at length two objections that were raised by philosophers of Aristotelian inspiration against the use of mechanical concepts in geometry. After discussing the first objection, that one cannot properly talk of the center of gravity of geometrical figures, Torricelli took up the second, to the effect that in reality the threads of a balance are never parallel but tend toward the center of the earth, by claiming that

> if this same balance, even though corporeal, were considered to be not on the earth's surface but in the highest regions beyond the sun's sphere, then the threads, while still drawn to the center of the earth, would be very much less convergent to each other, would be quasi-parallel. Let us imagine a mechanical balance transported beyond the starry balance [i.e., the constellation of that name] in the firmament, to an infinite distance. It will be understood by everybody that the suspension threads would no longer be convergent, but would be exactly parallel. . . . The Geometer has the special privilege to carry out, by abstraction, all constructions [*operationes*] by means of the intellect. Who, then, would wish to prevent me from freely considering figures hanging on a balance imagined to be at an infinite distance beyond the confines of the world?[21]

This passage should be interpreted as a declaration of the intellectual freedom, the "special privilege" granted to the geometer, who is not bound by physical constraints in the definition of his abstractions. Nothing could stop Torricelli

given examples of solids infinite in one or two dimensions whose volumes are finite, but had also explained that such results obtained because the infinite increase of one dimension is countered by the infinite decrease of another. See, among his other works that mention such figures, Nicole Oresme, *Le livre du ciel et du monde,* ed. Albert D. Menut and Alexander J. Denomy (Madison: Univ. Wisconsin Press, 1968), p. 119. However, Oresme's results were long forgotten by the early seventeenth century, and mathematicians and philosophers of the caliber of Roberval, Mersenne, Descartes, Christiaan Huygens, Fermat, and Barrow never challenged Torricelli's claim to priority.

[20] Marin Mersenne, *Universae geometriae mixtaeque mathematicae synopsis* (Paris, 1644), p. [2]. For Euclid's definition see *Elements,* bk. 11, def. 1.

[21] Torricelli, *De dimensione parabolae,* in *Opere,* Vol. I, p. 97; translation partially taken from Alexandre Koyré, *Galileo Studies,* trans. John Mepham (Atlantic Highlands, N.J.: Humanities Press, 1978), p. 243.

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from conceiving a figure, like the acute hyperbolic solid, that is so long as to extend "beyond the confines of the world."

Torricelli's claims on the nature of mathematical abstraction and the geometer's freedom to define his objects are consistent with various forms of Platonism and conceptualism.[22] But, as we shall see in the discussion of Barrow, the acceptance of Torricelli's solid presents potential difficulties for those philosophies of mathematics that ground geometrical existence in actual physical existence, for an infinitely long solid may turn out to be problematic for one's cosmological views. Consequently, Torricelli's solid may provide an incentive for separating mathematics from physics and may serve, as it did for Gassendi, as a clear indication that geometry is a self-contained discipline free of any material admixture.

In the rest of the article we shall study how some important philosophers active in the third quarter of the seventeenth century—Gassendi, Barrow, and Hobbes—reacted to the issues generated by Torricelli's cubature.[23] As one should expect, given their various interests and backgrounds, their reactions differed in intensity and in scope. For this reason we shall look at each philosopher individually, and our exposition will be chronological rather than thematic.

THE PHILOSOPHICAL REACTIONS

Pierre Gassendi

Although he was a habitual member of Mersenne's circle and was perhaps aware of Torricelli's result quite early, Gassendi did not use it in his discussion of mathematics until very late in his career. The first mention of the hyperbolic solid appeared in 1658 in his posthumous work *Syntagma philosophicum,* though a nearly identical discussion that made no reference to Torricelli's cubature appeared in the 1649 work *Animadversiones.*[24] Nonetheless, Torricelli's result became paradigmatic for Gassendi, for it very clearly showed the gulf between mathematics and physics that his philosophy theorized and also provided a powerful example of the value of geometry as an abstract science.

Like so many other empiricists, Gassendi held the traditional Aristotelian view that mathematics deals with quantity abstracted from matter and that mathematical concepts are obtained from sensation through a process of abstraction. In his

[22] On the delicate question of Torricelli's Platonism see Koyré, *Galileo Studies,* esp. p. 243; Paolo Galluzzi, "Evangelista Torricelli: Concezione della matematica e problema degli occhiali," *Annali dell'Istituto e Museo di Storia della Scienza di Firenze,* 1976, *1*:71–95; and Galluzzi, "Vecchie e nuove prospettive torricelliane," *La scuola galileiana: Prospettive di ricerca* (Florence: La Nuova Italia, 1979), pp. 13–51. Also of interest are the introduction by Luigi Belloni to the Italian translation of the selected works by Torricelli (*Opere scelte di Torricelli* [cit. n. 2]) and the remarks by Lucio Lombardo Radice in "La dissimulazione onesta," *Rinascita,* 1975, *48*:21–22.

[23] The reader may be surprised that two obvious candidates—Descartes and Spinoza—are not discussed. Descartes heard of Torricelli's result very early. In December 1643 he praised Roberval's proof in a letter to Mersenne: Mersenne, *Correspondance* (cit. n. 4), Vol. XII, pp. 377–378. But three years later (2 Nov. 1646), when his relations with Roberval had soured, he wrote Mersenne that Roberval's achievement, compared with Torricelli's, was minor: see *Oeuvres de Descartes,* ed. Charles Adam and Paul Tannery, 12 vols. (Paris: Vrin, 1976), Vol. IV, p. 553. However, although he held interesting views concerning our knowledge of infinity and Torricelli's result bears on them, Descartes did not comment on its philosophical implications. Similarly, in spite of his interest in the notion of infinity, as far as we know Spinoza mentions Torricelli only in connection with the void.

[24] Pierre Gassendi, *Syntagma philosophicum,* in *Petri Gassendi Opera omnia,* 6 vols., Vol. I–II (Lyon, 1658); rpt. (Stuttgart/Bad Canstatt: F. F. Verlag, 1964); and Gassendi, *Animadversiones in decimum librum Diogeni Laertii,* in *Opera omnia,* Vol. VI.

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lengthy *Rebuttals* to Descartes's *Meditations* he took issue with the view that we have innate ideas of mathematical objects. We learned about Euclid's triangle from teachers or books, he claimed, "or else we derived it by our own thinking as we reasoned about the triangle and other figures which were presented to our senses or that we had formed by putting together, breaking down, or moving about the ones that had been present in the senses."[25]

Although mathematical concepts originate in experience, Gassendi maintained a clear separation between mathematics and physics. While physicists are constrained by the material nature of the universe, mathematicians—and especially geometricians—free themselves through a process of abstraction from the "crassness, obstinancy and impediment of matter" and create a free universe in which, in contrast to the real one, there are no indivisible atoms but lines infinitely divisible into other lines, surfaces into other surfaces, and volumes into other volumes.

> And these are the suppositions from which Mathematicians, within the gates of pure and abstract Geometry and almost constituting a kingdom of their own, weave those famous demonstrations, some so extraordinary that they even exceed credibility [*fidem*], like what the famous Cavalieri and Torricelli showed [*ostenderunt*] of a certain acute solid infinitely long which nevertheless is equal to a parallelepiped or to a finite cylinder. It is clear, therefore, how they protect that kingdom of theirs in which they find [*excogitant*] so many remarkable and pleasurable things and take care not to mix anything with matter.[26]

This passage has been used by such influential readers as Pierre Bayle, Leon Brunschvicg, and Carl Boyer as evidence that Gassendi held skeptical views about mathematics. In note D to his article on Zeno of Sidon, Bayle enlisted Gassendi in his campaign against mathematics, and, after noting Gassendi's belief in a gulf between mathematics and physics, claimed that he gave "an example of the vanity of [the mathematicians'] would-be demonstrations, namely that two subtle Mathematicians allegedly have proved that a finite and an infinite quantity would be equal."[27] Bayle continued by citing from Gassendi's quotation the part in which Cavalieri and Torricelli are mentioned by name.

Given the paradoxical nature of Torricelli's result, Bayle was certainly correct in singling out the reference to it as the place to look for Gassendi's alleged skeptical views on mathematics. However, his reading was both imprecise and misleading. Contrary to Bayle's strong intimation, Gassendi did not say that Torricelli's result entailed the absurdity that the finite is equal to the infinite; rather, as we saw, he merely gave a vague but—as far as it went—correct statement of

[25] Gassendi, *Syntagma,* in *Opera omnia,* Vol. I, p. 264b; and Gassendi, *Metaphysical Colloquy, or Doubts and Rebuttals concerning the Metaphysics of René Descartes with His Replies,* in *The Selected Works of Pierre Gassendi,* ed. and trans. Craig B. Brush (New York: Johnson Reprint, 1972), pp. 256–257 (quotation).

[26] Gassendi, *Syntagma,* in *Opera omnia,* Vol. I, p. 264b.

[27] Pierre Bayle, *Dictionnaire historique et critique* (Paris: Edition Sociales, 1974), s.v. "Zenon." See also Leon Brunschvicg, *Les étapes de la philosophie mathematique* (Paris: Presses Universitaires de France, 1947), pp. 167–168; and Carl Boyer, *The History of the Calculus and Its Conceptual Development* (New York: Dover, 1959), p. 224. Both Brunschvicg and Boyer clearly base their views about the issue on Bayle's account. Bernard Rochot, in "Gassendi et les mathematiques," *Rev. Hist. Sci.,* 1957, *10*:69–77, says very little on the issue, which seems to have been unjustly neglected. It is probable that while Gassendi was a skeptic about mathematics in 1624, he had modified his position by 1649.

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Torricelli's and Cavalieri's theorems. Moreover, and more important, when Gassendi was presented with the paradoxical nature of Torricelli's cubature, far from launching into a tirade against the vanity of geometry, he made a concerted if perhaps unsuccessful attempt to give that science an honorable place in his system. For, after pointing out that not all geometrical theorems could be transported into physics, he suggested that once their different ontologies—of points and atoms, respectively—are made clear, geometry can be applied to physics because no sensible error would be generated. Furthermore, not only is geometry applicable, albeit with some caution, to physics, but it is also "a speculative science, which therefore does not concern itself with practical applications, but only has as its end the truth of its conclusions and the pleasure felt from the fact that, although extraordinary indeed, they follow with such evidence and certainty."[28] Geometry, then, is a coherent system of truths that is valuable on its own, independent of applications and is capable of generating pleasure, the *summum bonum* every human tries to attain.[29] Because of its paradoxical nature Torricelli's result, far from showing the vanity of geometry, points both to its independence from physics and to its structural coherence and beauty.

However, one cannot avoid a sense of dissatisfaction with Gassendi's treatment of Torricelli's cubature because of his silence about its bold infinitistic nature. The dissatisfaction mingles with surprise when one considers Gassendi's views on infinity. Failing to see how we could concoct the idea of infinity out of sensation, he maintained that we have a merely negative notion of infinity and criticized Descartes's claims in the Third Meditation about our understanding of God as infinite. He pointed out that "we call infinite that thing whose limits we have not perceived, and so by that word we do not signify what we understand about a thing, but rather what we do not understand." Nor does our inability to have a positive notion of infinity involve only ignorance of the infinite degree to which God possesses his perfections (i.e., what Gassendi called "infinity of perfection"); it also involves ignorance of more mundane sorts of infinity, such as infinite extension. In fact, he told Descartes that "infinitude either of place or of perfection cannot be understood."[30]

Since he maintained that we have no positive idea of infinity, one would expect Gassendi to attempt a finitistic reading of Torricelli's result; instead, he ignored the whole issue.[31] Gassendi's silence provided no solution to the challenge Torricelli's theorem presented to traditional empiricist views of our knowledge of infinity. Unless one were prepared to interpret Torricelli's result finitistically, one could hardly maintain after 1644 that we have no positive idea of infinity, for he offered a geometrical proof about an infinitely long solid. In this sense, we can

[28] Gassendi, *Syntagma,* in *Opera omnia,* Vol. I, p. 265b. Indeed, Gassendi himself used a mathematical approach to motion in *De proportione qua gravia decidentia accelerantur* (1646), in *Opera omnia,* Vol. III.

[29] *Ibid.,* Vol. II, pp. 693b, 695a.

[30] Gassendi, *Metaphysical Colloquy,* in *Selected Works,* ed. Brush (cit. n. 25), p. 212.

[31] Perhaps Gassendi's silence can be attributed to his views about space, which he thought of as per se void, isotropic, immovable, resistanceless, and infinite. Although some commentators (e.g., Edward Grant) have questioned that Gassendi viewed space as infinite, noting that at times he labeled it *immensum* rather than *infinitum,* he clearly identified immensity with infinite extension: "preater eandem concipimus infinitatem quesi extensionis, quae dicitur Immensitas"; see Gassendi, *Syntagma,* in *Opera omnia,* Vol. I, p. 190a. Perhaps, then, he did not attack the infinity of Torricelli's solid because any criticism of the infinity of the solid could be turned against his view of space.

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justifiably say that Gassendi failed to understand one of the important consequences of the cubature of the hyperbolic solid, or at least that he was unable to modify his empiricist epistemological approach in the light of Torricelli's result.

Isaac Barrow

Barrow's discussion of Torricelli's result and its implications occurred in lecture sixteen of his *Lectiones,* which was delivered in 1666 and was devoted to the topic of proportion. Torricelli's cubature is mentioned in connection with a discussion of the Aristotelian dictum that " 'there is no proportion between the finite and the infinite' . . . the truth of which statement, a very usual and well known [*pervulgatum*] axiom, seems to have been in part broken by the sagacity of modern geometricians, since they demonstrate a just proportion or even equality of innumerable planes and solids protracted to infinity with other finite planes and solids: which prodigy the very famous geometer Torricelli exhibited first." The relevance of the new theorems about infinitely long figures for Aristotle's maxim lies in the fact that they show that "not everything without limits in extension is *ipso facto* infinite in quantity; or yet more plainly . . . [that] from the infinity of one dimension that of a surface or a body does not follow."[32]

The fact that Barrow made such an explicit connection between Aristotle's dictum and Torricelli's result is at first a bit surprising. For although Aristotle's statement occurred in the middle of his arguments against the possibility of an infinite body, his discussion had been framed within a context that was much more physical than geometrical.[33] Furthermore, one would be hard pressed to find Aristotle's claim among the explicit principles of the classical mathematical tradition Barrow was well acquainted with. However, Barrow's readiness to discuss Aristotle's claim in the light of Torricelli's result is understandable once we comprehend that he, like many of his contemporaries (see, e.g., Hobbes's view), embraced the traditional Aristotelian view of mathematics not as an isolated and self-contained discipline, but as embedded in a web of other branches of knowledge with which it shared fundamental assumptions. Indeed, he went so far as to claim that geometrical axioms "are to be drawn from some higher and universal science, as first philosophy." Since Barrow claimed that Aristotle's dictum had been refuted only "in part," he proceeded to explain what had to be retained, namely, that "there is no proportion between a finite magnitude or quantity and an infinite quantity or magnitude of the same genus. But we are not to understand that in general there is no ratio between a limited figure and an unlimited magnitude."[34] In Barrow's eyes, then, Torricelli's cubature has the effect of disambiguating Aristotle's saying by rendering clear that it holds not indiscriminately, but only for magnitudes of the same genus, for example, for lengths with

[32] Isaac Barrow, *Mathematici professoris lectiones,* lect. 16, in *The Mathematical Works,* ed. W. Whewell (Cambridge: Cambridge Univ. Press, 1860), p. 255. For Aristotle's dictum see *De coelo* 1.6, 274a. Interestingly, it is probable that Galileo accepted Aristotle's dictum: see the first day of Galileo, *Two New Sciences,* trans. Drake (cit. n. 15), p. 42.

[33] So, e.g., Aristotle argues against an infinite body on the basis that it would be infinitely heavy: see *De coelo* 1.6.

[34] Barrow, *Mathematical Works,* Lect. 7, p. 107; Lect. 16, p. 256. Barrow held that some axioms are necessary to all reasoning and gave a few: e.g., that sense in its own way is an undoubted criterion for truth, that the possibility of a thing is argued from the existence of its like. See *ibid.,* Lect. 7, p. 110.

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lengths, areas with areas, and volumes with volumes. With this proviso Aristotle's dictum can be maintained, because it does not allow the intuitive, widespread, and erroneous inference from infinite length to infinite volume or area.

If Torricelli's cubature provided Barrow with an opportunity to clarify an important philosophical statement, it also presented him with serious problems concerning the nature of geometrical entities. He was opposed to any sort of view that posited the existence of mathematical entities only in the human, or even the divine, mind, for otherwise they would be mere "dreams or idols of things nowhere existing." Consequently, he adopted the strong position that

> all imaginable geometrical figures are really inherent in every particle of matter; I say really inherent in fact [*actu*] and to the utmost perfection, though not apparent to the senses, just as the effigy of Caesar lies under unhewn marble. . . . So, if the hand of an Angel (at least the power of God) should think to polish any particle of matter without vacuity, a spherical surface of a figure exactly round would appear to the eyes.[35]

Barrow's view that the reality of geometrical entities is grounded in their material existence is compatible with the mathematical existence of the infinite solid only if that solid is actually instantiated in some piece of matter. Barrow, then, like Aristotle, by linking geometry and cosmology raised, perhaps unwittingly, the issue of whether the latter had the resources to handle the former. Since he held that space is "really eternal and infinite," there is certainly room, as it were, in his cosmology for Torricelli's solid. The fact that he believed the amount of matter in the universe to be finite would seem to present a problem.[36] But of course one can see that this criticism involves applying Aristotle's dictum in the very indiscriminate fashion Barrow condemned so convincingly. For although Torricelli's solid is infinitely long, it has a finite volume and hence can be embedded, as it were, in a finite amount of matter. However, an infinitely long solid like Torricelli's requires an infinitely long piece of matter to be instantiated in, and although Barrow, unlike Aristotle, has the cosmological resources to build such a chunk of matter, there can hardly be an independent reason for holding that it really exists. One must conclude that although the mathematical existence of Torricelli's solid is compatible with this aspect of Barrow's view of geometrical existence, it strains it almost to the point of rupture.

Yet in addition to the position that geometrical entities must actually be instantiated, and possibly in competition with it, Barrow apparently held a weaker view, which one can gauge from what he said about demonstrations. All demonstrations, he claimed, proceed from true hypotheses. But in a circuitous way, as we shall see, he de facto equated the truth of a hypothesis with its self-consistency. This is why "demonstrations may be made of things which never had existence anywhere, nor will ever exist; for, it is sufficient for a demonstration to assume a true hypothesis, i.e. such as to imply no inconsistence in themselves." It is important to understand that for Barrow the requirement of self-consistency is not a self-contained logical point with no further purpose but the logical prerequisite for a metaphysical step that confirms the strict ties between the various branches of knowledge and their final subordination to metaphysics and theol-

[35] *Ibid.*, Lect. 5, pp. 84–85.
[36] *Ibid.*, Lect. 10, pp. 160 (quotation), 152.

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ogy. For Barrow claimed that "the truth of a hypothesis intimates the possible existence of the thing supposed; this possibility denotes the efficient cause of the thing (since otherwise it would be impossible for it to exist), and the efficient cause of all things is God." Since logical consistency is viewed as a mere prerequisite for possible instantiation, which conceptually involves the notion of divine production, Barrow could conclude, albeit fallaciously, that every demonstration supposes the existence of God as the creator not only of the knowing subject but also of "the knowable object."[37]

The consequence of this position is that even if a geometrical item is not actually instantiated, its geometrical existence is guaranteed if it could be instantiated. This is why "the dominion of reason far extends the limits of nature: the intelligible world is vastly more extended and diffusive than the sensible world and the acuteness of the mind contemplates many more things than the bodily senses."[38] Clearly, under this view of geometrical entities in which their existence is located in possible instantiation, the infinite length of Torricelli's solid poses no problem if one is ready to concede the nontrivial assumption that the very notion of an infinite solid is consistent. For, since space is infinite, God could certainly locate an infinitely long solid in it.

Barrow had a good understanding of Torricelli's result. Moreover, he clearly saw that the theorem can be intuitively explained by the fact that "the infinite diminution of one dimension compensates the infinite increase of the other," and he successfully used it to clarify Aristotle's dictum.[39] However, the same cannot be said about his grasp of the problems the hyperbolic solid presented for his philosophy of mathematics. In fact, Torricelli's solid helps to focus attention on an apparently unresolved tension in Barrow's theory. For although he consistently rejected the view that geometrical existence is grounded in human or divine understanding alone, he did not clarify his position as to whether it requires actual or merely possible material instantiation.

Thomas Hobbes

The most spirited and philosophically coherent discussion of Torricelli's solid was provided by Hobbes. It is difficult to determine when he was first exposed to Torricelli's result. We know that during his second stay in Paris in the 1640s he renewed his acquaintance with Mersenne and became a good friend of Gassendi's. It is possible—indeed probable, given the interest Torricelli's result generated—that he got some wind of it then. However, Hobbes's explicit treatment of the hyperbolic solid occurred much later, in the course of the controversy with John Wallis. The phases and various issues of this famous exchange are well known and need not be repeated here.[40] Still, one of its central aspects, concerning the role of infinity in mathematics, is directly relevant to our topic, and consequently we shall devote a few lines to it.

Wallis was a brilliant and outspoken supporter of the new mathematics who

[37] *Ibid.*, Lect. 7, pp. 110, 111 (quotations).

[38] *Ibid.*, p. 113.

[39] *Ibid.*, Lect. 16, p. 225. As we noted, Oresme had already pointed out that an infinitely long solid can have a finite volume if, while one dimension increases infinitely, the other decreases infinitely.

[40] On this controversy see John Scott, *The Mathematical Work of John Wallis* (New York: Chelsea, 1981), pp. 166–173.

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suggested that geometry is subordinate to arithmetic and attempted to introduce bold infinitistic considerations in both. In the dedicatory epistle of his most famous work, published in 1655 and appropriately entitled *Arithmetica infinitorum,* he claimed that he was picking up where Cavalieri (whom he knew only through Torricelli's work) had stopped. In the course of the work he studied infinite series by extending to them results obtained, often perilously, in finite series and tried to apply the arithmetical results thus obtained to the solution of geometrical problems. As a consequence he dealt, with little in the way of explanation, with infinitely long figures and "proved" some theorems about them. He was therefore able to claim in the scholium to proposition 107 that "that famous problem (indeed ingenious and much to be admired) which Torricelli exhibited in one solid figure (that is, construct a cylinder equal to the *Acute Hyperbolic Solid* continued to infinity), we have exhibited in other innumerable figures, both plane and solid."[41]

As much as Wallis was enthusiastically trying to legitimize infinity in mathematics, so much was Hobbes bent on repudiating it. Hobbes's views on infinity originate in his empiricist epistemology. He insisted that we have ideas only of what we sense or of what we can construct out of ideas so sensed; for example, we perceive horses and men, we form ideas of them, and therefore we can conceive of a centaur by compounding them. Since the ideas we obtain directly from experience are of finite things and repeated composition of such ideas cannot produce the idea of an infinite thing, Hobbes claimed that anything we conceive of must have a finite magnitude. Consequently, when we say that something is infinite, all we can justifiably mean is that "we are not able to conceive the ends, and bounds of the thing named; having no conception of the thing, but of our own inability."[42]

Since one of the main sources of error is the use of names for which we have no corresponding conception, it is not surprising that any consideration of infinity, both geometrical and arithmetical, ought to be banished. Hobbes claimed that when mathematicians use the word *infinite* what they usually mean, or ought to mean, is "indefinite," that is, as great, or small, as one pleases.[43] A mathematician who says "Let a line be extended to infinity" should not be understood as holding that such an operation could be carried out, but rather as allowing the reader to extend the line as long as he pleases. One might object, as Wallis did, that considering an infinite line does not entail extending it, since by hypothesis such a line cannot be extended by us, but merely supposing it extended, and consequently that Hobbes's attack misses the mark.[44] However, such an objec-

[41] John Wallis, *Arithmethica infinitorum,* in *Opera mathematica,* 2 vols. (Oxonii 1656–70, Vol. I, pp. 257 (dedicatory epistle), 412 (quotation).

[42] Thomas Hobbes, *Leviathan,* in *The English Works of Thomas Hobbes,* ed. William Molesworth (London, 1845), rpt. (Darmstadt: Scientia Verlag, 1966), Vol. III, pp. 6 (idea of a centaur), 17 (idea of infinity).

[43] Hobbes, *Opera philosophica quae Latine scripsit omnia,* ed. William Molesworth (London: J. Bohn, 1845); reprinted (Darmstadt: Scientia Verlag, 1966), Vol. V, p. 213. Hobbes's point was reasonable. André Taquet, in his *Dissertatio phisico-mathematica de motu circuli et sphaerae* (1650), in *Opera mathematica* (Antwerp, 1669), Vol. II, p. 146 states as a definition, the tenth, that "cum *infinitum* dicimus intellige indefinitum, sive quantumvis magnum, aut multum; sic enim loquuntur Geometrae."

[44] Hobbes, *Opera,* Vol. V, p. 154; and John Wallis, "Answer to Four Papers of Mr. Hobs, Lately Published in the Month of August, and This Present September, 1671," *Philosophical Transactions,* 1671, 75:2241–2250, esp. p. 2242.

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tion would be unfair, for from Hobbes's standpoint to suppose an infinite line as given would be begging the question, because the only way to gain a conception of an infinite line would be to extend it. Hence, actual infinity being ruled out, in Hobbes's mathematical universe every figure, at every stage of production, is finite. Consequently, to talk about a solid "*whose sides are extended out to infinity* is absurd. In fact, something is not called a chair before it is finished [*perficitur*]; so neither is something called a space before it is finished. But what is infinite is neither finished, nor can it be finished."[45]

Thus were the battle lines clearly drawn. In 1671 Hobbes wrote three letters to the Royal Society in which he attacked, among other of Wallis's contentions, the one that there are geometric figures of finite area or volume that have no center of gravity. He strongly suggested that to see the falsity of this claim one needed neither skill in geometry nor a knowledge of logic, but merely the common understanding all humans have. As usual, Wallis's reply was quick and caustic; in the same year he commented in the *Philosophical Transactions:*

> A surface, or solid, may be *supposed* so constituted, as to be *Infinitely Long,* but Finitely Great, (the Breadth Continually Decreasing in greater proportion than the Length increaseth) and so as to have no *Center of Gravity*. Such is *Torricellio's Solidum Hyperbolicum acutum;* and others innumerable, discovered by Dr. *Wallis,* Monsieur *Fermat,* and others. But to determine this, requires more *Geometry and Logic* than Mr. Hobs is Master of.

Wallis's reference to Torricelli clearly caught Hobbes by surprise. In his reply he claimed "I do not remember this of Torricellio, and I think Dr. Wallis does him wrong and Monsieur Fermat too. For, to understand this for sense, it is not required that a man should be a geometrician or a logician, but that he should be mad."[46] Such an answer was hardly satisfying. It looked more like a confession of ignorance than a proper reply to Wallis's sarcasm, and consciousness of the inadequacy of the reply may be what led Hobbes to the rather extensive treatment of Torricelli's solid in his last book-length mathematical work, *Principia et problemata aliquot geometrica ante desperata nunc breviter explicata et demonstrata,* published in 1672.

In the last chapter of that work, "On Infinity," Hobbes explained that the proper nonmathematical notion of infinity involves the idea of a being greater than any assignable measure. For example, an infinite space would be one that cannot be included by boundaries.

> But it has been demonstrated, they say, by Torricelli that a certain acute hyperbolic solid, in this sense of *infinite,* is equal to a cylinder whose basis has a diameter equal to half the base of the hyperbola and whose height is equal to the latus transversum of the hyperbola [*sic*]. I have read the demonstration of this problem quite often and with attention and have found no paralogism. However, I have found that the distance which Torricelli supposes *infinite* is to be understood as *indefinite*. Nor could it

[45] Hobbes, *Opera,* Vol. V, p. 209 (emphasis in original). Hobbes got so involved in the issue that he mentioned it in his autobiography in Latin verse: "Addidit Oxoniae Prealector Savilianus / Wallisius multo nobile dogma magis: / Nempe infinitae molis finem esse, et habere / Finitum partes et sine fine datas: / Quae duo fecerunt insanos dogmata, quotquot / festinaverunt esse geometrici." See *Thomae Hobbes Malmesburiensis vita Carmine expressa authore seipso* (1679), in *Opera,* Vol. I, p. xcv.

[46] Hobbes, *English Works,* Vol. VII, p. 433; Wallis, "Answer to Four Papers," p. 2243 (indeed, Torricelli's solid has no center of gravity); and Hobbes, *English Works,* Vol. VII, p. 445.

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> be understood differently by him, who in quite a lot of demonstrations uses Cavalieri's principle of *indivisibles,* which are such that their aggregate can be equated to whatsoever given magnitude. Therefore, a proposition as absurd as this, *that the infinite is equal to the finite,* must not be attributed to Torricelli. In act, as is clear by the natural light, there cannot be a solid so subtle which does not infinitely exceed every finite solid.[47]

This long passage shows the paradoxical power of Torricelli's result and how the bizarre properties of his solid entail a clear embarrassment for Hobbes's position on several counts. The theorem, as Torricelli proudly pointed out, is about an apparently infinitely long solid—bitter medicine for Hobbes, who viewed any consideration of infinity in mathematics as wholly pernicious. Moreover, the result was obtained geometrically, both by indivisibles and by exhaustion, according to a demonstrative methodology far removed from what Hobbes considered —not without justification—Wallis's rather cavalier use of infinity and algebraic manipulations. In other words, Torricelli had extended the realm of geometry well beyond the finitistic territory Hobbes was ready to grant it by employing methods that Hobbes himself could not but accept. Furthermore, Hobbes believed that natural light teaches us that an infinitely long solid, however subtle, must exceed in volume any finite solid. Nor is the reference to natural light merely occasional or at the periphery of Hobbes's mathematical views, for it was his considered opinion that the principles of knowledge include all the propositions that are known by natural light and that all the principles of mathematics are known to be true by natural light.[48] Since, then, an infinitely long solid must have an infinite volume, he could conclude that interpreting Torricelli's result as saying what it seems to say entails the absurdity that the infinite is equal to the finite. No wonder, then, that he read the demonstration "quite often and with attention," perhaps in the vain hope of finding an error.

But the proof is correct, and the only avenue left open for Hobbes was to try to understand Torricelli's result as dealing not with an infinitely long solid but with an indefinitely long one, and consequently to reinterpret the theorem in finitistic terms. It is unfortunate that Hobbes did not explain how the finitistic reading of the result is supposed to go, but presumably what he had in mind is this. The hyperbolic solid, viewed as indefinitely long, can be elongated as much as one pleases and, consequently, its value can be made to differ from that of the cylinder by less than any given quantity. This is certainly true, but hardly satisfactory, for each of the successive hyperbolic solids thus generated is finitely long but not equal in volume to the cylinder. Of course one could construct an infinite succession of volumes of such solids and claim, by a limit process, that the succession converges to the volume of the cylinder. However, attributing such a view to Hobbes would not only be historically inaccurate, because it would involve forcing our own mathematical notions on him, but would also foist on him conclusions he would hardly tolerate. In fact, although Hobbes would certainly accept the hyperbolic solids whose volumes are elements of the succession, he would have to reject the passage to the limit, since the limit of the succession does not

[47] Hobbes, *Opera,* Vol. V, p. 213. Hobbes is not very careful here; he misstates the theorem by confusing the height of the cylinder with the diameter of its base. However, the error is trivial and without consequence for what follows.

[48] *Ibid.,* pp. 155, 157.

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belong to the succession itself, and in order to be equal to the volume of the cylinder the limit must be equal to the volume of an actually infinitely long solid. In sum, Torricelli's result is boldly infinitistic, and Hobbes's attempts at reducing it to a finitistic framework are destined to fail.

Not only did Hobbes claim that Torricelli's theorem should be given a finitistic reading; he also claimed that Torricelli himself, who used Cavalieri's methods, must, or should, have understood it so. This is a strange claim indeed. Torricelli certainly interpreted his result in infinitistic terms; furthermore, Cavalieri too, perhaps spurred by Torricelli's cubature, proved by indivisibles that a certain solid infinite length has finite volume. However, Hobbes's views on Cavalieri's indivisibles may help to dispel some of the bafflement.

In accordance with his empiricist beliefs, Hobbes held that a mathematical line is a body of which we consider only the length and that the very notion of a line without breadth is inconceivable and deleteriously used in geometry. As a consequence, limiting our attention to planes, he considered their indivisibles not as segments, but as very minute parallelograms. Furthermore, given his denial of infinity, he viewed them not as infinitely small (the result of infinite division), but merely as undivided. Since indivisibles are neither one-dimensional nor infinitesimals, if added together, or multiplied by a finite number, they can exceed any surface.[49] In sum, in Hobbes's view, Cavalieri's indivisibles are Archimedean quantities that have a ratio to whatever given (finite) surface according to the Eudoxian definition in Euclid's *Elements*. Presumably the same, *mutatis mutandis,* is true of indivisibles of solid figures, which he must have viewed as very thin solids.[50]

But then, since natural light teaches us that an infinitely long solid cannot have a finite volume, the indivisibles composing it (all of them finite) could not have a Eudoxian ratio to it. Hence Torricelli must, or should, have viewed his solid not as infinitely but as merely indefinitely long. Of course, there is a sense in which the argument we have attributed to Hobbes begs the question, since it assumes what Wallis and Torricelli rejected, namely, that an infinitely long solid must have a infinite volume. But Hobbes's assumption is not an unreasonable one, and he can hardly be faulted for adhering to it.

Hobbes's ill repute as a mathematician is even greater than his good repute as a political philosopher; he is heralded as a genius of the first order in the latter capacity and as an ungifted and presumptuous dilettante in the former. There is some truth in this severe judgment. His contributions to mathematical discoveries are limited at best, and his mistakes, which were many, are far from being interesting or inspiring to mathematicians. Even his mathematical analysis of Torricelli's cubature is finally ill conceived and fails to do justice to the striking originality of the theorem. However, Hobbes was sufficiently philosophically alert to grapple with issues, such as those arising from Torricelli's result, that were at the forefront of contemporary mathematical debate and to come to grips with advanced problems and methods. He saw that Torricelli's cubature could be used to support the new brand of modern mathematics heralded by Wallis, a

[49] *Ibid.,* Vol. IV, p. 58, Vol. V, p. 206 (on lines); and Vol. IV, pp. 74, 173–174, 75 (on the indivisibles of planes).

[50] Euclid, *Elements,* Bk. 5, def. 4. In his view of the indivisibles of solid figures, Hobbes was not far from what Torricelli himself seems to have thought; see note 10.

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mathematics that, in Hobbes's view, would lead into the quagmire of infinity and finally undermine the certainty of geometry, "the only Science it hath pleased God hitherto to bestow on mankind."[51]

CONCLUSION

Torricelli's hyperbolic solid was the first of a flurry of geometric figures infinite in one or even two dimensions that were found and studied by early-modern mathematicians. As such, it stretched the intuitive universe of geometrical figures constituting the background of the seventeenth-century geometer and generated problems for views that connected geometrical and material existence too tightly. Moreover, Torricelli's cubature of an infinitely long solid employed not only heuristic—or at least questionable—methods, such as indivisibles or Wallisian infinite series, but demonstrative procedures that could hardly be rejected. The hyperbolic solid in essence first revealed the inadequacy of the empiricist notion of infinity for seventeenth-century mathematical developments. Later in the century John Locke conceded that inadequacy when he concluded a long discussion of the empiricist view that we have a negative idea of infinity by admitting, without further elaboration, that mathematicians might have other ways of obtaining ideas of infinity.[52] This capitulation, like the reactions discussed above, indicates that Torricelli's result deserves a place not only in the history of seventeenth-century mathematics but also in the history of philosophy of the period.

[51] Hobbes, *Leviathan,* in *English Works,* Vol. III, pp. 23–24.

[52] John Locke, *An Essay Concerning Human Understanding,* ed. Peter H. Nidditch (Oxford: Clarendon Press, 1975), Bk. 2, Ch. 17, sect. 22, p. 223. For reasons too complex to be discussed here, after the third quarter of the seventeenth century the fortune of Torricelli's solid began to decline. However, it found its way into the mathematical dictionaries of the eary eighteenth century and even into the *Encyclopédie.* See John Harris, *Lexicon techicum* (London: 1704), s.u. "hyperbolicum acutum"; Edmund Stone, *A New Mathematical Dictionary* (London: Senex, 1726), s.v. "hyperbolicum acutum"; and *Encyclopédie; ou, Dictionnaire raisonnée des Sciences, des arts et des métiers* (math. ed. d'Alembert) (Lausanne/Berne: 1751–1780) (rpt. Stuttgart–Bad Canstatt: Fromann, 1966), s.v. "solide hyperbolique aigu." Indeed, it can still be found nowadays, sometimes under the colorful name of Gabriel's horn, in calculus textbooks.

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